

Realidades Virtual e Aumentada

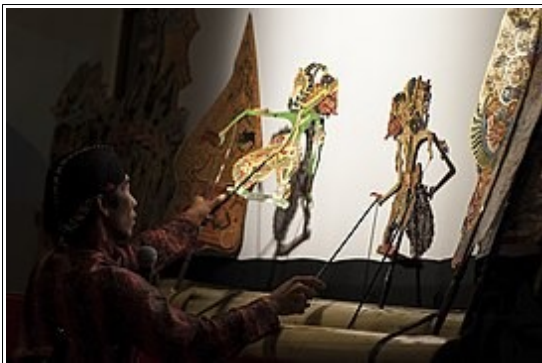
2ª Aula | 2024.09.26 | Prof. André Sier

Enquadramento artístico de Realidades Virtual (RV), Realidade Mista (FX) e Aumentada (RA).

Introdução à Realidade Virtual. Tecnologias de imersão primitivas (Marionetas de Sombras, Zootrópios, Daguerreótipos, Dioramas). Tecnologias de imersão contemporâneas e análise das obras *TheSwordofDamocles*, *TheLegibleCity*, *LiveTapedVideoCorridor*, *TVBuddha*, *Wolfenstein3d/Doom*, *Videoplace*, *TheWatching*, *Osmose*, *TheNightJourney*, *TheArtvertiser*, *Levelhead*, *Chalkroom*.

Tecnologias de imersão primitivas

https://en.wikipedia.org/wiki/Shadow_play



wayang, an Indonesian shadow puppet form

Shadow play, also known as **shadow puppetry**, is an ancient form of storytelling and entertainment which uses flat articulated cut-out figures (shadow puppets) which are held between a source of light and a translucent screen or scrim. The cut-out shapes of the puppets sometimes include translucent color or other types of detailing. Various effects can be achieved by moving both the puppets and the light source. A talented puppeteer can make the figures appear to walk, dance, fight, nod and laugh.

Plato's Cave and the Stubborn Persistence of Ignorance

Plato's cave metaphor illustrates the cognitive trap of ignorance, where we may be unaware of the limitations of our understanding.



By: Daniel R. DeNicola

The most memorable image of ignorance occurs in what is probably the most famous passage of all philosophy: Plato's Allegory of the Cave in "The Republic." Recall the scenario: human beings dwelling in the darkness of an underground cavern, bound at the legs and neck so that they cannot move, even to turn their heads. They have no other memory of life, since they have been imprisoned in this way since childhood. Before them, they see only moving shadows that are cast by objects unknown to them, illumined by a flickering fire that we are told lies somewhere behind them. They know nothing of this except the shadows and hear only echoes from the voices of their keepers, whom they have never seen. In such a benighted state, they pass their days.

ShadowPlayVR: Understanding Traditional Shadow Puppetry Performance Techniques Through Non-Intuitive Embodied Interactions.

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[VRST '23: Proceedings of the 29th ACM Symposium on Virtual Reality Software and Technology](#)

Article No.: 73, Pages 1 - 2 <https://doi.org/10.1145/3611659.3617226>

Abstract

"ShadowPlayVR" is a virtual reality system designed to introduce the intricate art of Chinese shadow puppetry into Virtual Reality (VR), focusing on the non-intuitive embodied interactions that emulate puppetry performance. By incorporating an immersive, experiential learning approach, ShadowPlayVR offers users a hands-on understanding of this art form. Preliminary testing reveals the significant role of contextual information in facilitating understanding and mastering these complex interactions. The work also presented showcases how VR can serve as a powerful tool to preserve and engage with traditional cultural heritage in a contemporary digital context.

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«Recently, Viveiros de Castro (2014) has shown us that Amazonian and other Amerindian populations live in conceptual universes where plants, animals, spirits are all conceived as persons, where boundaries dilute between the human and the non-human, and the latter is even higher praised and revered as an inspirational model. He also goes on to acclaim, as Caillois, the anthropological importance of masks and devices worn by the shamans, both at personal and social levels, as a token interface that bridges the human to the underworld (Kerényi 2008, Caillois 1990), a process taking place and performatively intensified towards a completed transformation onto the otherness it seeks to incorporate and blend into, ultimately blending with Gaia, as seen at 1.1.1, or with a particular manifestation of one of its aspects, as a non-human animal or as a mythological being, drawing its insights from all the non-human communication seemingly taking place: a subject of utmost significance that has lead the construction of special VR sculptural mask interfaces for the immersive works *Babylon* (2018) and *0X Labyrinth* (2018-2022)»

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<https://andre-sier.com/wolfanddotcom/0xlabyrinth/>

Zoetrope <https://en.wikipedia.org/wiki/Zoetrope>



A **zoetrope** is a pre-film animation device that produces the illusion of motion, by displaying a sequence of drawings or photographs showing progressive phases of that motion. A zoetrope is a cylindrical variant of the phénakisticope, an apparatus suggested after the stroboscopic discs were introduced in 1833. The definitive version of the zoetrope, with replaceable film picture film strips, was introduced as a toy by Milton Bradley in 1866 and became very successful. The name *zoetrope* was composed from the Greek root words *ζωή* *zoe*, "life" and *τρόπος* *tropos*, "turning" as a translation of "wheel of life". The term *soetrope* was coined by inventor William E. Lincoln.[1]

Earlier rotating devices with images



The Mysteries of Nature and Art (1635)

An earthenware bowl from Iran, over 5000 years old, could be considered a predecessor of the zoetrope. This bowl is decorated in a series of images portraying a goat jumping toward a tree and eating its leaves.[2] Though the images are sequential and seem evenly distributed around the bowl, to have the images appear as an animation the bowl would have to rotate quite fast and steadily while a stroboscopic effect would somehow have to be generated. As such, it remains very uncertain if the artist who created the bowl actually intended to create an animation.

<https://en.wikipedia.org/wiki/Daguerreotype>



Still life with plaster casts, made by Daguerre in 1837, the earliest reliably dated daguerreotype[[note 1](#)]

Daguerreotype (/dəˈɡɛər(i.)ə, -(i.)oʊ-/ ⓘ;[1][2] French: *daguerréotype*) was the first publicly available photographic process, widely used during the 1840s and 1850s. "Daguerreotype" also refers to an image created through this process.

Invented by Louis Daguerre and introduced worldwide in 1839,[3][4][5] the daguerreotype was almost completely superseded by 1856 with new, less expensive processes, such as ambrotype (collodion process), that yield more readily viewable images. There has been a revival of the daguerreotype since the late 20th century by a small number of photographers interested in making artistic use of early photographic processes.[6]

The first authenticated image of Abraham Lincoln, a daguerreotype of him as U.S. Congressman-elect in 1846, attributed to Nicholas H. Shepard

To make the image, a daguerreotypist polished a sheet of silver-plated copper to a mirror finish; treated it with fumes that made its surface light-sensitive; exposed it in a camera for as long as was judged to be necessary, which could be as little as a few seconds for brightly sunlit subjects or much longer with less intense lighting; made the resulting latent image on it visible by fuming it with mercury vapor; removed its sensitivity to light by liquid chemical treatment; rinsed and dried it; and then sealed the easily marred result behind glass in a protective enclosure.

Camera obscura

The *camera obscura* (Latin for "dark chamber") in its simplest form is a naturally occurring phenomenon.[58]

A broad-leaved tree in bright sunshine will provide conditions that fulfill the requirements of a pinhole camera or a camera obscura: a bright light source (the sun), the shade that the leafy canopy provides, a flat surface onto which the image is projected and holes formed by the gaps between the leaves. The sun's image will show as a round disc, and, in a partial eclipse, as a crescent.[59]

A clear description of a camera obscura is given by Leonardo da Vinci in Codex Atlanticus (1502): (he called it *oculus artificialis* which means "the artificial eye")[60]



If the facade of a building, or a place, or a landscape is illuminated by the sun and a small hole is drilled in the wall of a room in a building facing this, which is not directly lighted by the sun, then all objects illuminated by the sun will send their images through this aperture and will appear, upside down, on the wall facing the hole. In another notebook, he wrote:



camerae obscurae and plates for daguerreotype called "Grand Photographe" produced by [Charles Chevalier](#) ([Musée des Arts et Métiers](#))

You will catch these pictures on a piece of white paper, which placed vertically in the room not far from that opening, and you will see all the above-mentioned objects on this paper in their natural shapes or colors, but they will appear smaller and upside down, on account of crossing of the rays at that aperture. If these pictures originate from a place which is illuminated by the sun, they will appear colored on the paper exactly as they are. The paper should be very thin and must be viewed from the back.

[61]

In the 16th century, Daniele Barbaro suggested replacing the small hole with a larger hole and an old man's spectacle lens (a biconvex lens for correcting long-sightedness), which produced a much brighter and sharper image.[\[58\]](#)[\[62\]](#)[\[63\]](#)[\[64\]](#)

By the late 18th century, small, easily portable box-form units equipped with a simple lens, an internal mirror, and a ground glass screen had become popular among affluent amateurs for making sketches of landscapes and architecture. The camera was pointed at the scene and steadied, a sheet of thin paper was placed on top of the ground glass, then a pencil or pen could be used to trace over the image projected from within. The beautiful but fugitive little light-paintings on the screen inspired several people to seek some way of capturing them more completely and effectively—and automatically—by means of chemistry.

Daguerre, a skilled professional artist, was familiar with the *camera obscura* as an aid for establishing correct proportion and perspective, sometimes very useful when planning out the celebrated theatrical scene backdrops he painted and the even larger ultra-realistic panoramas he exhibited in his popular Diorama.

Diorama



Monpa people at the Jawaharlal Nehru



Museum, Itanagar, Arunachal Pradesh, India

Exhibition Lab's mountain gorilla diorama at the American Museum of Natural History in New York City.

A **diorama** is a replica of a scene, typically a three-dimensional model either full-sized or miniature. Sometimes it is enclosed in a glass showcase for a museum. Dioramas are often built by hobbyists as part of related hobbies such as military vehicle modeling, miniature figure modeling, or aircraft modeling.^[*citation needed*]

In the United States around 1950 and onward, natural history dioramas in museums became less fashionable, leading to many being removed, dismantled or destroyed.

EtymologyThe word "diorama" originated in 1823 as a type of picture-viewing device, from the French in 1822. The word literally means "through that which is seen", from the Greek di- "through" + orama "that which is seen, a sight". The diorama was invented by Louis Daguerre and Charles Marie Bouton, first exhibited in Paris in July 1822 and at The Diorama, Regent's Park on September 29, 1823.^[*citation needed*] The meaning "small-scale replica of a scene, etc." is from 1902.^[2]

Daguerre's and Bouton's diorama consisted of a piece of material painted on both sides. When illuminated from the front, the scene would be shown in one state and

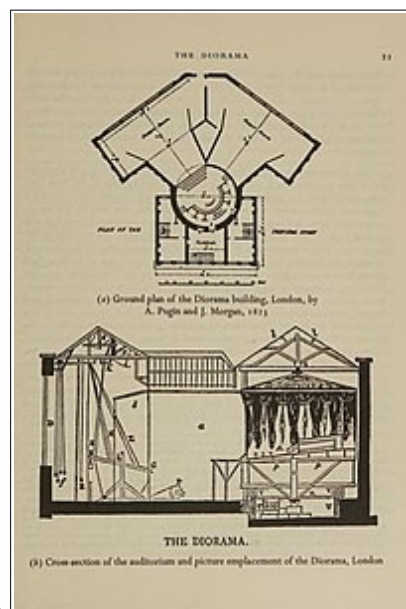
by switching to illumination from behind another phase or aspect would be seen. Scenes in daylight changed to moonlight, a train travelling on a track would crash, or an earthquake would be shown in before and after pictures.

Modern



Cooling tower construction diorama. The current, popular understanding of the term "diorama" denotes a partially three-dimensional, full-size replica or scale model of a landscape typically showing historical events, nature scenes or cityscapes, for purposes of education or entertainment. One of the first uses of dioramas in a museum was in Stockholm, Sweden, where the Biological Museum opened in 1893. It had several dioramas, over three floors. They were also implemented by the Grigore Antipa National Museum of Natural History from Bucharest Romania and constituted a source of inspiration for many important museums in the world (such as the American Museum of Natural History in New York and the Great Oceanographic Museum in Berlin) [reference below].

Historic



Daguerre and Bouton **Diorama Building, London 1823, by A. Pugin and J. Morgan (illustration reproduced from Gernsheim 1968, p 21)**

The Diorama was a popular entertainment that originated in Paris in 1822. An alternative to the also popular "Panorama" (panoramic painting), the Diorama was a theatrical experience viewed by an audience in a highly specialized theatre. As many as 350 patrons would file in to

view a landscape painting that would change its appearance both subtly and dramatically. Most would stand, though limited seating was provided. The show lasted 10 to 15 minutes, after which time the entire audience (on a massive turntable) would rotate to view a second painting. Later models of the Diorama theater even held a third painting.

The size of the proscenium was 24 feet (7.3 m) wide by 21 feet (6.4 m) high (7.3 meters x 6.4 meters). Each scene was hand-painted on linen, which was made transparent in selected areas. A series of these multi-layered, linen panels were arranged in a deep, truncated tunnel, then illuminated by sunlight re-directed via skylights, screens, shutters, and colored blinds. Depending on the direction and intensity of the skillfully manipulated light, the scene would appear to change. The effect was so subtle and finely rendered that both critics and the public were astounded, believing they were looking at a natural scene.

The inventors and proprietors of the Diorama were Charles-Marie Bouton (1781– 1853), a Troubadour painter who also worked at the Panorama under Pierre Prévost, and Louis Jacques Mandé Daguerre (1787–1851), formerly a decorator, manufacturer of mirrors, painter of Panoramas, and designer and painter of theatrical stage illusions. Daguerre would later co-invent the daguerreotype, the first widely used method of photography.

Tecnologias de imersão contemporâneas e análise das

obras *TheSwordofDamocles*, *TheLegibleCity*, *LiveTapedVideoCorridor*, *TVBuddha*, *Wolfenstein3d/Doom*, *Videoplace*, *TheWatching*, *Osmose*, *GoldenCalf*, *TheNightJourney*, *TheArtvertiser*, *Levelhead*, *GazingBall*, *Chalkroom*.